The · cat · sat · on · the · ___

MODERN AI FOR BEGINNERS · VIDEO 1

How AI Actually Thinks

AI vs ML vs DL vs Gen AI, and how ChatGPT works

[Your Name]

Every single word is a guess.

How does a guessing machine write poems, code and essays?

THE COURSE

Five steps to building with AI

01 · TODAY

Understand

How AI thinks

02

Prompt

Talk to it well

03

Build

Call AI from code

04

RAG

Connect your own data

05

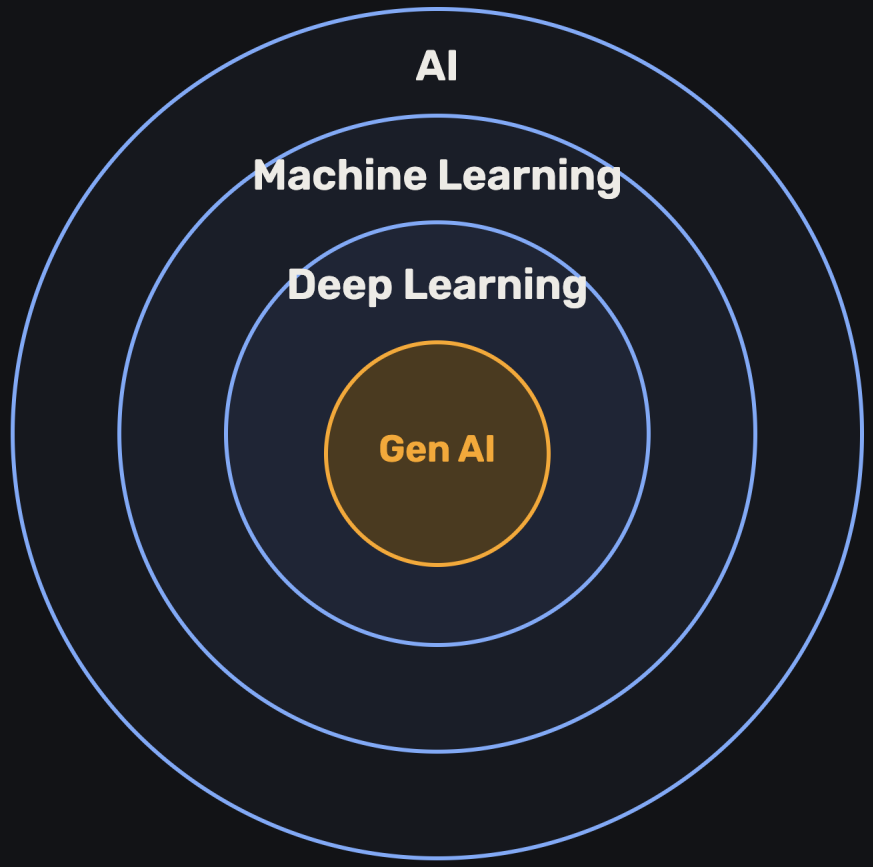
Agents

Give AI tools

Today is the foundation. Everything later sits on top of it.

PART 1 · THE AI FAMILY

Four words, one family



AI: machines acting smart

Machine Learning: learns from examples

Deep Learning: many-layered neural networks

Generative AI: creates new content

They fit inside each other, like Russian dolls.

ARTIFICIAL INTELLIGENCE

Any machine acting smart

It doesn't have to learn. Early AI was rules written by hand.

```
IF email has "lottery" AND "winner" → SPAM
IF email has "free money" → SPAM
```

Scammer writes "l0ttery" with a zero → the rule breaks.
Humans can't write a rule for everything.

Examples: old chess computers, game characters, "Press 1 for billing" phone menus

MACHINE LEARNING

Learn the rules from examples

Traditional programming	**Rules + Data → Answers**
Machine Learning	**Data + Answers → Rules**

You already use it every day:

- Netflix and YouTube recommendations
- Bank fraud alerts
- Swiggy and Zomato delivery times

The catch: humans still chose which clues to look at ("feature engineering").

DEEP LEARNING

Layers that find their own clues



Nobody told it to look for ears. It worked that out by itself.

Needs huge data and GPUs, so it took off around 2012.
Examples: Face unlock, Google Lens, voice assistants, Google Translate

GENERATIVE AI

From recognising to creating

Traditional AI recognises	Generative AI creates
"Is this a cat?" → Yes	"Draw a cat on the moon"
"Is this review positive?" → Yes	"Write the review"
"Is this code buggy?" → Yes	"Write the code"

Text: ChatGPT, Claude, Gemini · Images: Midjourney, DALL·E · Music: Suno · Video: Sora, Veo · Code: GitHub Copilot

PART 1 RECAP

The family in four lines

Term	In one line
AI	Machines acting smart, even with plain rules
Machine Learning	Learns the rules from examples
Deep Learning	Many-layered networks find their own clues
Generative AI	Creates new text, images, music, video, code

The most famous Gen AI today: the LLM, the tech behind ChatGPT.

PART 2 · HOW CHATGPT WORKS

LLM: Large Language Model

Large

Billions of tiny "knobs" called parameters. GPT-3 had 175 billion.

Language

Works with human language, and with code.

Model

A mathematical program that has learned patterns from text.

ChatGPT, Claude, Gemini and Llama are all LLMs, or built on them.

THE BIG IDEA

It predicts the next word

The capital of France is ___



It picks a word, adds it to the sentence, and predicts again.

ONE WORD AT A TIME

Guess, add, repeat

```
The cat sat on the → mat

The cat sat on the mat → and

The cat sat on the mat and → purred

The cat sat on the mat and purred → .

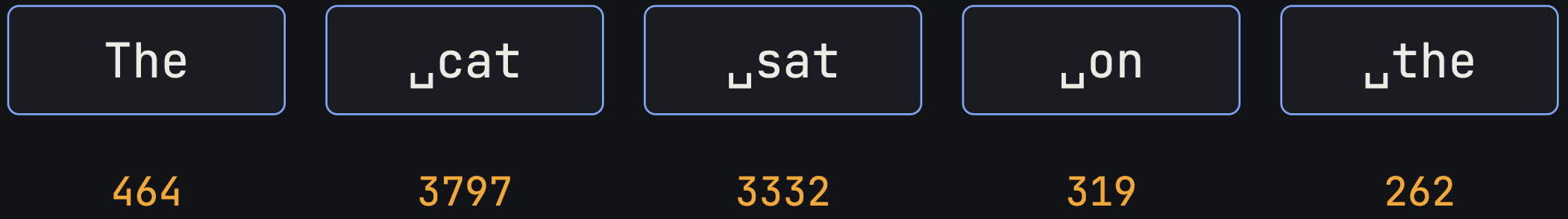
… until it predicts [END]
```

That's why answers appear word by word.

TOKENS

Text is chopped into pieces

"The cat sat on the"



Computers only understand numbers, so each piece gets an ID. ␣ = the space is part of the token.

1 token ≈ ¾ of an English word

TOKENS

Why tokens matter to you

You pay per token

AI APIs charge for every token you send and get back.

Languages differ

Hindi or Tamil often need more tokens than English for the same meaning.

"Strawberry"

It sees chunks like [straw] [berry], not letters, so counting R's is hard.

Live demo: paste the same sentence in English and your mother tongue into a tokenizer.

CONTEXT WINDOW

The model's desk

EVERYTHING MUST FIT ON THE DESK

- Your question
- The conversation so far
- Any document you paste in
- The answer it's writing

100K–1M+

tokens fit on today's desks. That's hundreds of pages.

Too much? The oldest parts fall off and are forgotten.

Remember this desk. It's the problem RAG solves.

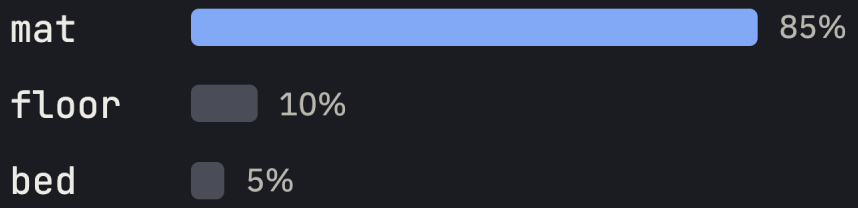
TEMPERATURE

A weighted dice roll

LOW · temperature 0

Safe and consistent

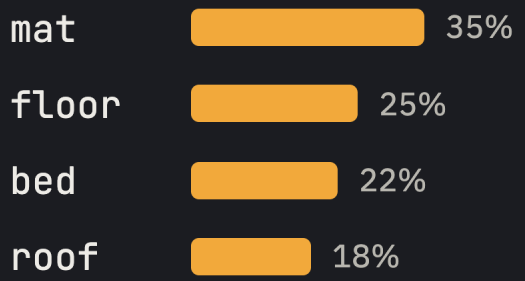
Facts, code, data extraction



HIGH · temperature 1.2

Creative and surprising

Poems, stories, brainstorming



Same question twice → different answers. The dice rolled differently.

HOW IT LEARNED

Three stages of training

Stage 1

Pre-training

Reads trillions of words. Plays one game billions of times: guess the hidden next word.

= reading the whole library

Stage 2

Fine-tuning

Trained on example conversations to follow instructions like an assistant.

= job training

Stage 3

Human feedback

People rank its answers. It learns to give more of the preferred ones.

= performance reviews

Thousands of GPUs · weeks to months · millions of dollars

When you chat, it isn't learning.

The knobs are frozen. Its knowledge stops at a training cutoff.

Memory and web search are tools added around the model, not the model learning.

LIMITS

Three big weaknesses

Knowledge cutoff

Doesn't know recent events.

No private data

Never read your documents, notes or manuals.

Hallucination

Trained to sound right, not to be right.

The fix, later in this course: RAG

RECAP

What you learned today

Idea	In one line
AI family	AI ⊃ ML ⊃ DL ⊃ Gen AI
LLM	Predicts the next word, one at a time
Tokens	Pieces of text, and what you pay for
Context window	The model's desk; overflow is forgotten
Temperature	Safe and consistent vs creative
Training	Pre-training, fine-tuning, human feedback
Limits	Cutoff, no private data, hallucination

HOMEWORK · 10 MINUTES

Try it yourself

Task 1 · Tokens

Count your tokens

Open the OpenAI tokenizer. Type one sentence in English and in your mother tongue. Which used more tokens?

Task 2 · Temperature

Ask twice

Ask any chatbot "Write a two-line poem about chai" twice. Why are the answers different?

Share your answers in the comments.

NEXT VIDEO

Prompt Engineering

The AI predicts from what's on its desk.
Next, learn to fill that desk well.